Question			Answer	Marks	AO	Guidance
5	(a)		**DR** $\frac{dy}{dx} = -2x + 32x^{-3} = 0$ $x^4 = 16$	**M1**	**1.1a**	Attempt to differentiate and equate to zero soi
			So $x = \pm 2$	**A1**	**1.1b**	Both x-values and no others.
			When $x = \pm 2, y = 7$ [So the points are $(-2, 7)$ and $(2, 7)$]	**A1**	**1.1b**	FT their x-coordinate(s) Do not FT $x = 0$
				[3]		
5	(b)		**DR** $\frac{d^2y}{dx^2} = -2 - 96x^{-4}$	**M1**	**2.1**	Attempts to find the second derivative. FT their $\frac{dy}{dx}$
			When $x = 2, \frac{d^2y}{dx^2} = -2 - \frac{96}{16} < 0$ When $x = -2, \frac{d^2y}{dx^2} = -2 - \frac{96}{16} < 0$			Or convincing statement that $\frac{d^2y}{dx^2} < 0$ for any $x[\neq 0]$ because x^{-4} is always positive
			So both points are maximum points	**E1**	**2.1**	**AG** Complete argument required from correct second derivative and their $x \neq 0$. ISW if $-2 - \frac{96}{x^4}$ wrongly evaluated. Also allow for one point established and an argument from symmetry
			Alternative method $\frac{dy}{dx} > 0$ for $0 < x < 2$ and $\frac{dy}{dx} < 0$ for $x > 2$ and $\frac{dy}{dx} > 0$ for $x < -2$ and $\frac{dy}{dx} < 0$ for $-2 < x < 0$	**M1**		Evaluating gradient for suitable values of x on either side of each turning point Also allow for y-coordinates in these ranges.
			So both points are maximum points	**E1**		AG Complete argument required from correct first derivative and their $x \neq 0$. Also allow for one point established and an argument from symmetry
				[2]		